Color Edge Detection in Presence of Gaussian Noise Using Nonlinear Prefiltering

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***Abstract*—A new technique for edge detection in color images corrupted by Gaussian noise is presented. The proposed method adopts a multipass processing approach that gradually reduces the noise in the R, G, and B components of the image. The prefiltering steps are specifically designed to operate in conjunction with the edge detection algorithm. They adopt two different models for data smoothing that aim at avoiding false edges produced by noise and at preserving the image details during noise removal. The subsequent algorithm for edge detection has been designed to further decrease the sensitivity to noise of the overall method. Thus, accurate edge maps can be achieved even in the presence of highly corrupted data. Results of computer simulations show that the proposed approach significantly improves our previous methods and performs better than other techniques in the literature.**

***Index Terms*—Edge detection, fuzzy systems, Gaussian noise, image processing, nonlinear filters.**

I. Introduction

With respect to gray-scale pictures, color images generally include a richer measurement information that can be successfully exploited in order to improve the performance of image-based instrumentation and/or extend its application range [1], [2]. In this framework, edge detection plays a very relevant role in the realization of a complete image understanding system. Indeed, high-level processing tasks such as image segmentation and object recognition directly depend on the quality of the edge detection procedure. It is well known, however, that the generation of an accurate edge map becomes a very critical issue when the images are corrupted by noise. In this respect, noise having Gaussian-like distribution is very often encountered during image acquisition [3].

Different approaches to color edge detection in presence of this kind of noise have been proposed in the scientific literature [4], [5]. Methods that represent extensions of monochrome edge detectors typically process the three color channels independently and combine the resulting edge maps. As an example, let us consider the application of the well-known Sobel technique to the three image channels. An edge point in the color image could be estimated by evaluating the maximum of the gradient components or their vector sum. It should be observed that the processing of three color channels is generally necessary to yield an accurate edge map. Indeed, edges in a color image can be produced by objects having the same (or similar) luminance but different chrominance information. Vector space approaches take into account this issue by modeling each image pixel as a three-dimensional vector in the assigned color coordinate system [6]. These methods generally adopt gradient-based algorithms that resort to an appropriate definition of a distance in the given color space. An interesting example is represented by the family of directional operators that aim at detecting the location and orientation of edges in color images [7]. Another effective class of edge detection techniques is based on order statistics [8]. Basically these operators are characterized by linear combinations of rank-ordered pixel vectors. Operators with different performance and efficiency can be designed by appropriately choosing the set of coefficients in the linear combination. *Difference vector* (DV) methods are another poweful family of edge detectors: a gradient is typically computed in each of the four main directions and the maximum gradient vector is selected to detect edges [4]. Since gradient-based methods are very sensitive to noise, subfiltering or prefiltering algorithms are generally adopted [9]. However, the performance of such methods is limited, especially when the image data are highly corrupted. As a result, the errors in the produced edge map may become very annoying.

In this paper a novel method for color edge detection in Gaussian noise is presented. The proposed approach adopts a multipass processing architecture that aims at reducing noise before extracting the image edges. The data smoothing has been specifically designed to work in combination with the edge detection process. For this purpose, we developed a two model-based prefiltering scheme that aims at achieving the following goals: accurate preservation of image details and reduction of false edges caused by noise. With respect to our previous single-model filtering technique [10], the proposed approach gives a more appropriate smoothing behavior because it considers the effects of different classes of noisy pixels on the subsequent edge detection procedure. In order to further increase the accuracy of the method, additional filtering is embedded in the algorithm for edge detection. A well-balanced compromise between noise smoothing and spatial resolution has been carefully taken into account in the design of this stage. The improvements with respect to our previous methods are significant. As a result, the proposed architecture outperforms state-of-the-art techniques in the literature.

This paper is organized as follows. Section II focuses on two model-based prefiltering, Section III presents the noise-protected edge detector, Section IV discusses the validation of the method, and Section V reports conclusions.

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II. Image Prefiltering for Edge Detection

The data prefiltering for edge detection is based on the classification of noisy pixels into two different (fuzzy) classes:

1) pixels corrupted by noise with amplitude not too different from that of the neighbors (type A pixels);
2) pixels corrupted by noise with amplitude much larger than that of the neighbors (type B pixels).

Let us focus on type B pixels that are typically outliers, i.e., large amplitude noisy pixels present in the data as an effect of the "tail" of the Gaussian distribution. Since their probability is low, their number is limited. However, their effect can become very annoying, because the presence of outliers in the filtered data produces false edges and then reduces the quality of the edge map. For this reason, two different filtering techniques have been adopted in this paper to deal with type A and type B pixels, respectively.

Let us suppose we deal with digitized multichannel RGB images. As mentioned above, the proposed method is based on multipass processing. Thus, let $X^{(p)}$ briefly denote the multichannnel image at the pass p, where $p = 0$ denotes the input noisy image. Let $x_k^{(p)}(i,j)$ be the pixel value $\left(0 \le x_k^{(p)} \le L-1\right)$ at location $[i,j]$ in the kth channel $X_k^{(p)}$, where the R (red), G (green), and B (blue) channels are, respectively, denoted by $k=1$, $k=2$, and $k=3$. For a 24-bit color image, we have $L = 256$. The proposed multipass processing involves the following operations.

A. Type A Prefiltering

Type A prefiltering takes into account the differences between the pixel to be processed and its neighbors as follows: small differences are considered noise to be reduced; large differences are considered edges to be preserved. A two-step procedure is applied to the image channels in order to increase the effectiveness of the smoothing action (Fig. 1). This procedure is defined by the following relationships:

$$x_k^{(1)}(i,j) = x_k^{(0)}(i,j) + \frac{1}{8}\sum_{\substack{m=-1\\(m,n)\neq(0,0)}}^{1}\sum_{n=-1}^{1}\zeta_k^{(1)}\left(x_k^{(0)}(i+m,j+n), x_k^{(0)}(i,j)\right) \quad (1)$$

$$x_k^{(2)}(i,j) = x_k^{(1)}(i,j) + \frac{1}{8}\sum_{\substack{m=-1\\(m,n)\neq(0,0)}}^{1}\sum_{n=-1}^{1}\zeta_k^{(2)}\left(x_k^{(1)}(i+m,j+n), x_k^{(1)}(i,j)\right) \quad (2)$$

where $\zeta_k^{(p)}$ is a parameterized nonlinear function

$$\zeta_k^{(p)}(u,v) = \begin{cases} u-v & |u-v| \le a_k^{(p)} \\ \left(\frac{3a_k^{(p)}-u+v}{2}\right)\mathrm{sgm}(u-v) & a_k^{(p)} < |u-v| \le 3a_k^{(p)} \\ 0 & |u-v| > 3a_k^{(p)} \end{cases} \quad (3)$$

and $a_k^{(p)}$ is an integer $0 < a_k^{(p)} < L$ $(k = 1,2,3)$. The choice of this function is based on our previous research work in the field of gray-scale images [10]. According to (1), the filtering action is first applied to the three input color channels $X_1^{(0)}$, $X_2^{(0)}$, and $X_3^{(0)}$. As a result, three intermediate filtered components $X_1^{(1)}$, $X_2^{(1)}$, and $X_3^{(1)}$ are obtained (see Fig. 1). A second filtering pass is thus applied to these data (see (2)) yielding the filtered components $X_1^{(2)}$, $X_2^{(2)}$, and $X_3^{(2)}$, respectively. Basically, the filtering algorithm considers the pixels that belong to a 3×3 window and gradually excludes the values that are very different from the central element, in order to avoid detail blur during noise cancellation. In more detail, when all the (absolute) differences between the central pixel and its neighbors are smaller than $a_k^{(p)}$, only noise is considered to be present. Thus, a strong smoothing is performed and the result is the arithmetic mean of the pixel values in the neighborhood. Differences larger than $3a_k^{(p)}$ clearly denote an image edge and then their contribution is zero [see (3)]. Intermediate situations (absolute differences larger than $a_k^{(p)}$ and smaller than $3a_k^{(p)}$) are processed as a compromise between these opposite effects.

According to the above considerations, this filtering action depends on the value of parameter $a_k^{(p)}$ only. Small values better preserve fine details; large values produce a strong noise cancellation. The choice of a satisfactory set of parameter values is based on our recently introduced technique for automatic parameter tuning [10]. Basically, this method varies the value of the parameter $a_k^{(p)}$ from a minimum to a maximum, and takes into account the *progressive mean square error* $\Delta\mathrm{MSE}\left(a_k^{(p)}\right)$ between the noisy image filtered with $a_k^{(p)}$ and the same image filtered with $a_k^{(p)}-1$. When the ΔMSE is maximum, a satisfactory estimate of the optimal $a_{k\,\mathrm{opt}}^{(p)}$ can be obtained. New experimental results have shown that this method can successfully be applied to the R, G, and B channels of real color images, even if the contrast of these components is generally lower with respect to gray-scale pictures.

B. Type B Prefiltering

Type B prefiltering takes into account the differences between the pixel to be processed and its neighbors in a different way: if all these differences are very large, the pixel is (possibly) an outlier to be cancelled. For this reason, the noise removal procedure adopts a different nonlinear model. It is briefly summarized as follows (see Fig. 1):

$$x_k^{(3)}(i,j) = x_k^{(2)}(i,j) - (L-1)\Delta(i,j) \quad (4)$$

where

$$\Delta(i,j) = \underset{\substack{m=-1,0,1\\n=-1,0,1\\(m,n)\neq(0,0)}}{\mathrm{MIN}}\left\{\mu_{\mathrm{LA}}\left(x_k^{(2)}(i,j), x_k^{(2)}(i+m,j+n)\right)\right\} - \underset{\substack{m=-1,0,1\\n=-1,0,1\\(m,n)\neq(0,0)}}{\mathrm{MIN}}\left\{\mu_{\mathrm{LA}}\left(x_k^{(2)}(i+m,j+n), x_k^{(2)}(i,j)\right)\right\} \quad (5)$$

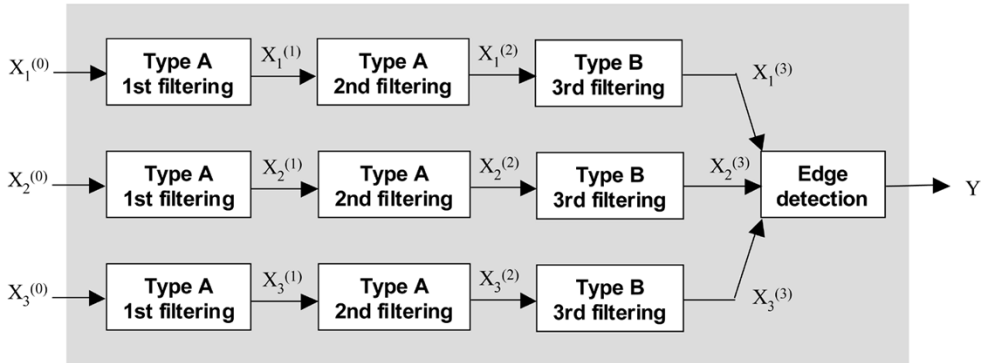


Fig. 1. Block diagram of the multipass processing.

and $\mu_{\mathrm{LA}}(u,v)$ denotes the membership function that describes the fuzzy relation: "u is much larger than v"

$$\mu_{\mathrm{LA}}(u,v) = \begin{cases} \frac{u-v}{L-1} & 0 < u-v \leq L-1 \\ 0 & u-v \leq 0. \end{cases} \tag{6}$$

As an example, let us consider a uniform neighborhood $x_k^{(2)}(i+m, j+n) = c\,(m=-1,0,1; n=-1,0,1; (m,n) \neq (0,0))$ and let $x_k^{(2)}(i,j) \gg c$ be a positive outlier. According to (5) we have $\Delta(i,j) = \left[x_k^{(2)}(i,j) - c\right]/(L-1)$. Thus (4) yields $x_k^{(3)}\,(i,j) = c$. The outlier has been removed. It should be observed that this filtering operation also increases the robustness of the proposed edge detector in presence of noise pulses.

III. Noise-Protected Edge Detector

Edge detection is performed using another class of fuzzy models. Even if the adoption of nonlinear prefiltering significantly reduces the Gaussian noise, the resulting image data are not noise-free. Thus, an edge detector with very low sensitivity to noise is necessary. For this purpose, we deal with color vectors that are obtained by performing the average of selected color pixels. The output of the color edge detector is given by the following relations:

$$y(i,j) = (L-1)[1 - \mathrm{MIN}\{\mu_{\mathrm{SM}}(B_1), \mu_{\mathrm{SM}}(B_2)\}] \tag{7}$$

$$B_h = \|\mathbf{A}_0, \mathbf{A}_h\|; \quad \mathbf{A}_h = \frac{1}{N_h} \sum_{\mathbf{v}_r \epsilon S_h} \mathbf{v}_r$$
$$h = 0,1,2 \tag{8}$$

where μ_{SM} is the membership function of fuzzy set "small," symbol $\|\|$ denotes the Euclidean distance, and S_h is a set of N_h color vectors $\mathbf{v}_h$. Suitable choices are

$$S_1 = \{\mathbf{x}^{(3)}(i-1,j-1), \mathbf{x}^{(3)}(i-1,j), \mathbf{x}^{(3)}(i-1,j+1)\}$$
$$S_2 = \{\mathbf{x}^{(3)}(i-1,j-1), \mathbf{x}^{(3)}(i,j-1), \mathbf{x}^{(3)}(i+1,j-1)\}$$
$$S_0 = \left\{\mathbf{x}^{(3)}(i,j), \mathbf{x}^{(3)}(i,j+1), \mathbf{x}^{(3)}(i+1,j), \mathbf{A}_1, \mathbf{A}_2\right\}.$$

These choices aim at realizing a satisfactory compromise between resolution and noise attenuation.

As an example, a larger number of elements in S_1, S_2, and S_0 would decrease the noise sensitivity at the price of a decrease of the spatial resolution. The performance depends on an appropriate selection of pixel patterns too. The above choices are based on a heuristic approach and intensive experimentation with image data. The membership function μ_{SM} is defined by the following relation:

$$\mu_{\mathrm{SM}}(u) = \begin{cases} 1 & 0 \leq u < b_1 \\ \exp\left[-\frac{(u-b_1)^2}{b_2^2}\right] & u \geq b_1 \end{cases} \tag{9}$$

where b_1 and b_2 are integer parameters. The choice of parameter values is not very critical. Basically, smaller values perform a stronger activation of the operator in the presence of image edges at the price of an increase of the sensitivity to noise (typically $b_1 = 5$, $30 < b_2 < 60$). The edge detection procedure aims at representing object contours as bright lines in the resulting edge map. Conversely, uniform regions are represented as dark areas. As an example, let us consider a perfectly uniform area. Since $B_1 = 0$ and $B_2 = 0$, we have $\mu_{\mathrm{SM}}(B_1) = 1$, $\mu_{\mathrm{SM}}(B_2) = 1$, and the result given by the edge detection procedure is a zero luminance value [see (7)]. It should be observed that the key difference with respect to our previous algorithm for color edge detection [11] is the computation of the vectors $\mathbf{A}_0$, $\mathbf{A}_1$, and $\mathbf{A}_2$. This new processing plays an important role in reducing the noise sensitivity of the operator.

IV. Validation of the Method

It is known that accurate detection of edges in noisy data should comply with two possibly conflicting requirements. The edge detection process should avoid false edges produced by noise and ensure that actual edges are correctly detected. In order to validate the performance of our edge detection method, we have performed many computer simulations dealing with synthetic and real images. An example of a synthetic test picture is shown in Fig. 2. The corresponding R, G, and B components are represented in Fig. 2(b)–(d), respectively. The RGB coordinates of the two colors are (200, 45, 90) and (60, 100, 180). Since both colors have the same luminance ($Y = 0.3\,\mathrm{R} + 0.6\,\mathrm{G} + 0.1\,\mathrm{B} = 96$), no object is perceivable in the corresponding luminance image and only a method that processes multichannel data can detect the edges. We generated a noisy test picture [Fig. 3(a)] by adding zero-mean Gaussian noise with standard deviation $\sigma = 12$ (The noise was generated using the algorithm in [12].) The result given by the proposed edge detector is shown in Fig. 3(b).

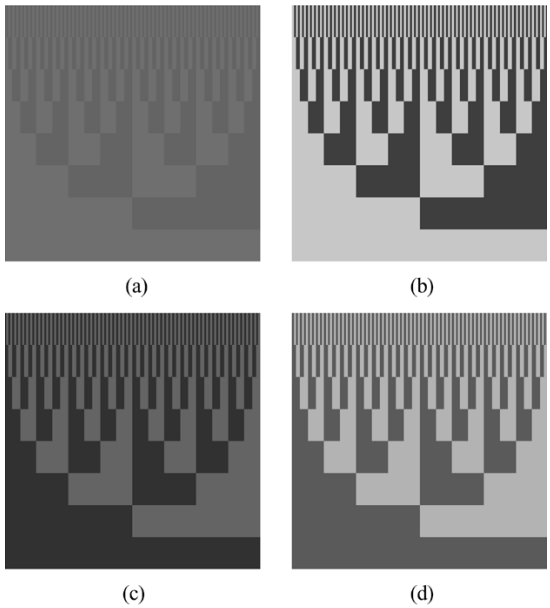


Fig. 2. (a) 24-bit RGB color synthetic image, (b) R component, (c) G component, and (d) B component.

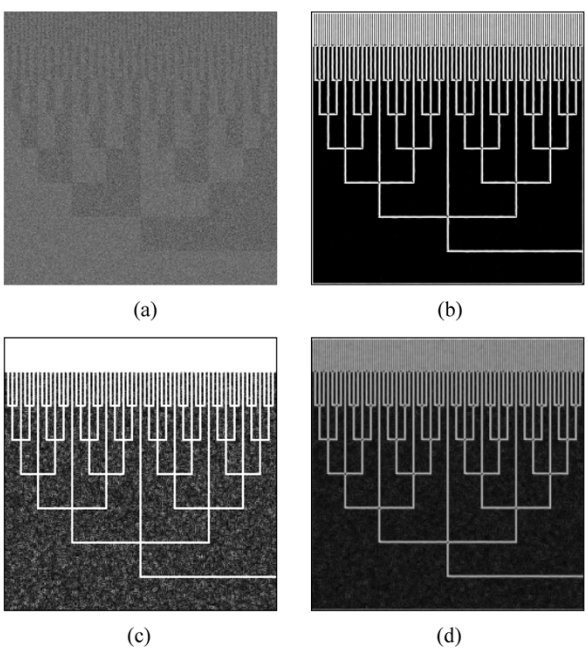


Fig. 3. (a) Noisy synthetic image. Results given by (b) the proposed method, (c) the Sobel operator, and (d) the DV-ANN edge detector.

For a comparison, we considered the edge map yielded by the Sobel operator [Fig. 3(c)] and the result of the application of the difference vector edge detector with adaptive nearest neighbor prefiltering (DV-ANN) [Fig. 3(d)]. The former is a well-known and widely adopted technique; the latter can be considered a state-of-the-art operator for color edge detection, according to the analysis reported in [4]. The lower resolution yielded by the Sobel technique is clearly perceivable observing Fig. 3(c). The thinnest vertical lines have not been detected. Moreover, the edge map is noisy, and this effect is very annoying. The DV-ANN detector [Fig. 3(d)] can extract these lines and the

Fig. 4. 24-bit RGB color real image.

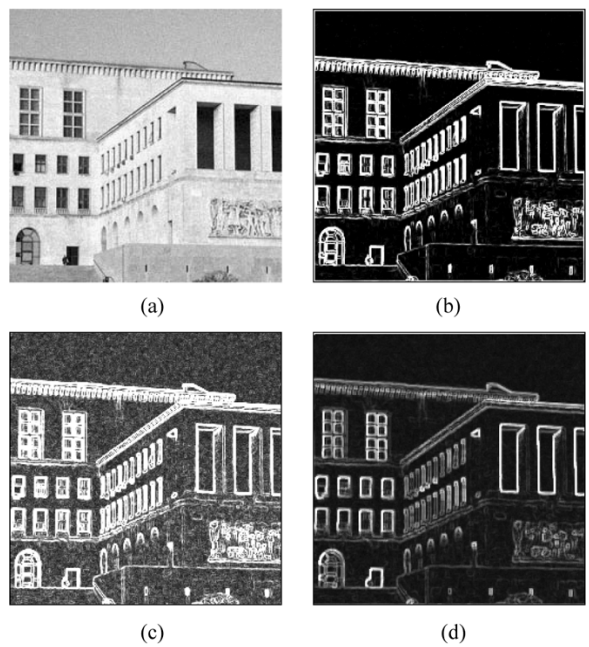
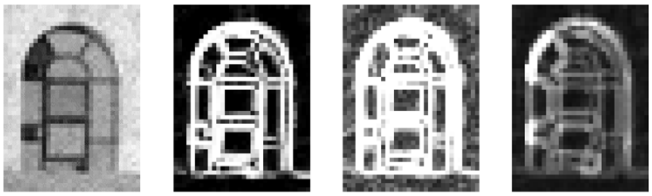


Fig. 5. (a) Noisy real image. Results given by (b) the proposed method, (c) the Sobel operator, and (d) the DV-ANN edge detector.

result is less noisy. However, the price to be paid is the blurring of the edge contours.

The proposed method [Fig. 3(b)] offers the best result among them. The thinnest vertical lines have satisfactorily been detected, the resulting edge map is much less noisy, and the image edges look very sharp. An example of a 24-bit real image is depicted in Fig. 4. The noisy image, obtained by adding zero-mean Gaussian noise with standard deviation $\sigma = 10$, is shown in Fig. 5(a). The results of the application of the mentioned edge detectors are depicted in Fig. 5(b) (proposed method), (c) (Sobel operator), and (d) (DV-ANN technique). The better performance of the proposed approach is apparent. In order to highlight the filtering and edge detection behaviors of the different methods, a detail of the image is also reported in Fig. 6. We can easily see that the edge map given by our technique [Fig. 6(b)] looks sharper and richer in detail than those yielded by the Sobel operator [Fig. 6(c)] and the DV-ANN method [Fig. 6(d)].

For a comparison, the results of the application of the mentioned edge detectors to the original noise-free data are reported

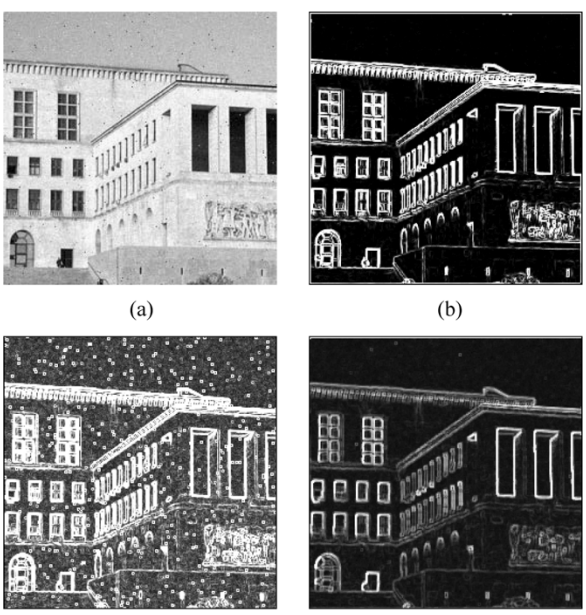


(a) (b) (c) (d)

Fig. 6. (a) Detail of the noisy image. Results given by (b) the proposed method, (c) the Sobel operator, and (d) the DV-ANN edge detector.

(a) (b) (c) (d)

Fig. 7. (a) Detail of the original noise-free image. Results given by (b) the proposed method, (c) the Sobel operator, and (d) the DV-ANN edge detector.



(a) (b) (c) (d)

Fig. 8. (a) Image corrupted by Gaussian and impulse noise. Results given by (b) the proposed method, (c) the Sobel operator, and (d) the DV-ANN edge detector.

in Fig. 7. The sensitivity to outliers can be investigated by further superimposing impulse noise with low probability (about 1%) to the noisy picture [Fig. 8(a)]. We can observe that our filtering scheme [Fig. 8(b)] yields a very effective noise cancellation without blurring the image details. Since almost all the outliers have been removed, no reduction of the quality of the edge map is perceivable with respect to the previous case. On the contrary, the sensitivity to outliers of the Sobel technique is very high [Fig. 8(c)]. Many false edges are clearly recognizable, and the result is very annoying. The sensitivity to outliers of the DV-ANN method is much lower and only a few false edges are produced.

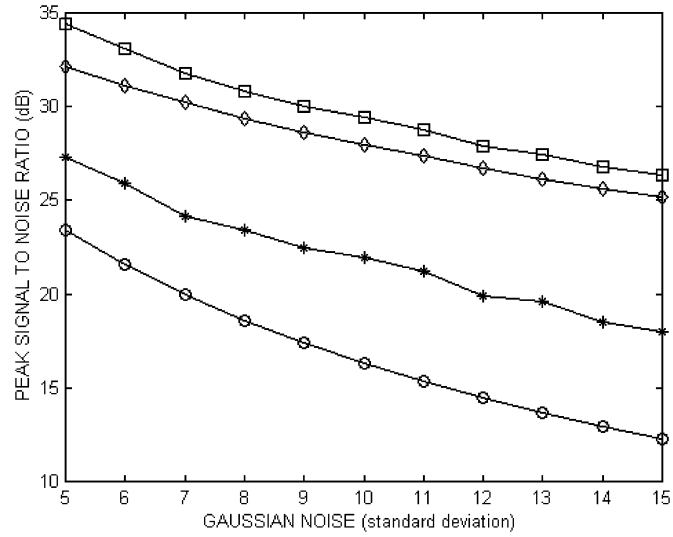


Fig. 9. PSNR values for image data corrupted by Gaussian noise. Plot of the results given by the proposed method (squares), the Sobel operator (circles), the DV-ANN edge detector (diamonds), and the application of our previous algorithms (asterisks).

In order to obtain a quantitative evaluation of the noise performance, we performed three groups of tests. In the first group we considered a set of 11 pictures corrupted by Gaussian noise with σ ranging from five to 15. We measured the noise performance by adopting the method proposed in [4], where the edge maps of the images corrupted by noise are compared with the edge maps of the original image for each edge detector. The corresponding noise performance is measured by means of the well-known peak signal-to-noise ratio (PSNR), which is defined as

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{\sum_i\sum_j (L-1)^2}{\sum_i\sum_j (y(i,j)-z(i,j))^2}\right) \tag{10}$$

where $y(i,j)$ is the pixel luminance at location $[i,j]$ in the edge map of the noisy image and $z(i,j)$ denotes the corresponding pixel luminance in the edge map of the original picture. A plot of the PSNR values given by the different methods is reported in Fig. 9. In order to highlight the effect of the novelties introduced in the proposed architecture, we considered for a comparison the results yielded by our previous (not noise-protected) edge detection algorithm with type A prefiltering. We can easily observe that the performance improvement given by the proposed method is very relevant (at least 7 dB with respect to our previous techniques). The new processing architecture including the noise protected edge detection algorithm and type A-B prefiltering significantly outperforms the DV-ANN method and the other techniques.

We performed a second group of tests by superimposing impulse noise with probability 1% to the previous set of noisy pictures. The corresponding plot of the PSNR values is shown in Fig. 10. We can see that when the images are corrupted by Gaussian noise and noise pulses, the performance of the Sobel technique and our previous method significantly decreases. Conversely, the results given by the proposed approach are very satisfactory and are better than those yielded by the DV-ANN

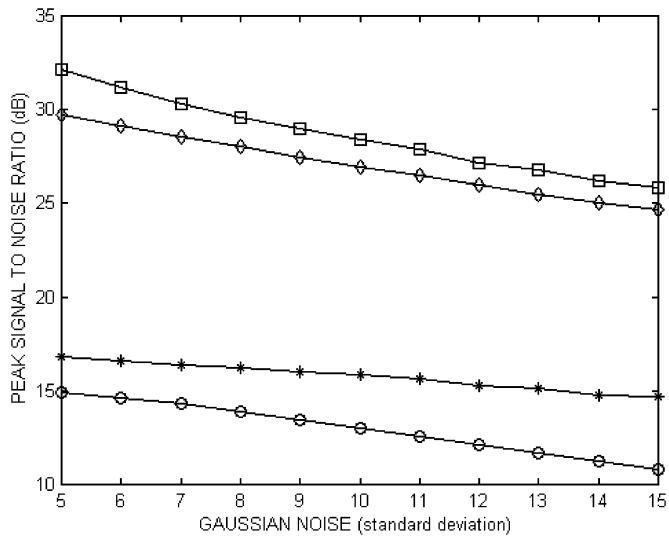


Fig. 10. PSNR values for image data corrupted by Gaussian and impulse noise. Plot of the results given by the proposed method (squares), the Sobel operator (circles), the DV-ANN edge detector (diamonds), and the application of our previous algorithms (asterisks).

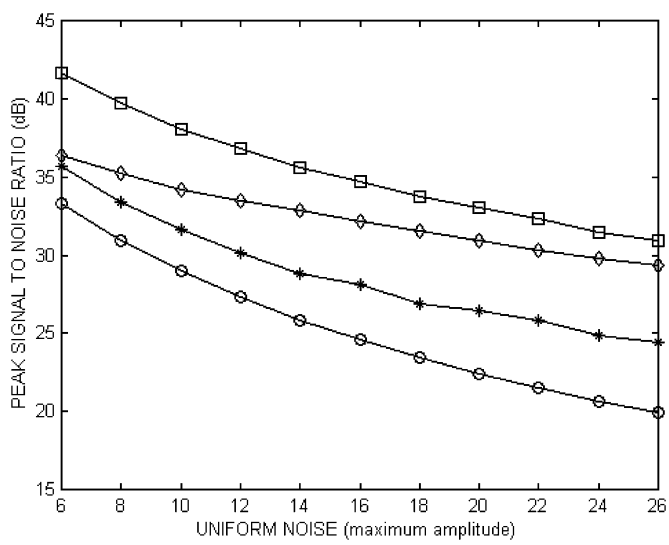
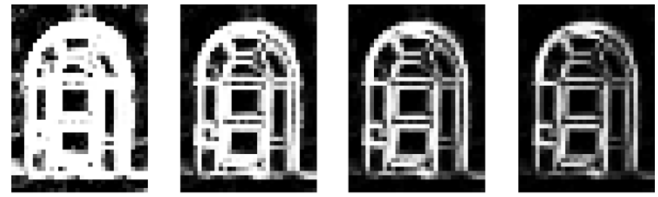


Fig. 11. PSNR values for image data corrupted by uniformly distributed noise. Plot of the results given by the proposed method (squares), the Sobel operator (circles), the DV-ANN edge detector (diamonds), and the application of our previous algorithms (asterisks).

method. Finally, we performed a third group of tests dealing with uniform noise. For this purpose, we generated a set of 11 noisy pictures by adding uniformly distributed noise with maximum amplitude ranging from 6 to 26. The plot of the PSNR values given by the different methods is depicted in Fig. 11. The better performance of the proposed method is apparent. In the above groups of tests we adopted the same parameter settings for our edge detection algorithm: $b_1 = 5$ and $b_2 = 55$. The satisfactory results obtained with different kinds of noise corruption show that the choice of parameter values is not critical in our approach. In general, small values of b_1 and b_2 increase the sensitivity of the edge detector to fine details and to noise as well. For a given application, a satisfactory compromise between these effects can be easily found.



(a) (b) (c) (d)

Fig. 12. Effects of different choices of b_2 $(b_1 = 5)$: (a) $b_2 = 10$, (b) $b_2 = 30$, (c) $b_2 = 50$, (d) $b_2 = 70$.

(a) (b) (c) (d)

Fig. 13. Effects of different choices of b_1 $(b_2 = 40)$: (a) $b_1 = 0$, (b) $b_1 = 10$, (c) $b_1 = 20$, (d) $b_1 = 30$.

As an example, let us consider the noisy image in Fig. 6(a). The effects of different values of b_2 are shown in Fig. 12 for a given value of b_1. The sensitivity of the edge detector is too high when $b_2 = 10$ [Fig. 12(a)] and too low when $b_2 = 70$ [Fig. 12(d)]. More satisfactory results can be found for $b_2 = 30$ [Fig. 12(b)] or $b_2 = 50$ [Fig. 12(c)]. The effects of different values of b_1 are depicted in Fig. 13 for a given value of b_2. Suitable choices are $b_1 = 0$ [Fig. 13(a)] and $b_1 = 10$ [Fig. 13(b)]. Too large values such as $b_1 = 20$ [Fig. 13(c)] and $b_1 = 30$ [Fig. 13(d)] produce missing edges and should be avoided.

V. Conclusion

A new method for edge detection in color images has been presented. The proposed approach deals with data corrupted by Gaussian noise and adopts a novel multipass prefiltering scheme in order to increase the accuracy of the edge detection process. This filtering section is based on the combination of nonlinear models that improves and extends our previous research work to RGB color image processing. The new edge detection algorithm operates in the color vector space and has been specifically designed to further reduce the errors caused by Gaussian noise (possibly) still affecting the data after prefiltering.

Computer simulations dealing with synthetic and real images have shown that the quality of the resulting edge map is very satisfactory and that the method yields more accurate results than using other techniques in the literature.

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